

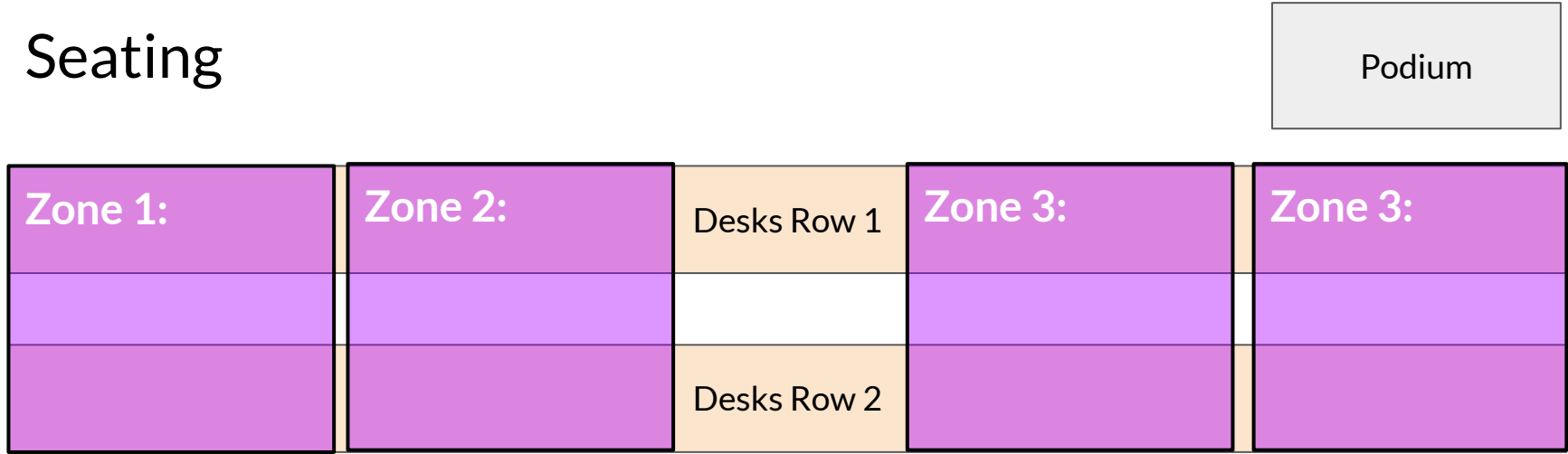
CSE 151B: Deep Learning

Instructor: Annapurna Vadaparty

TA: Ben Bao

Tutors: Alexander Radulescu, Trevor Duong

Seating



Start talking to the people in your zone:

1. What was the hardest part of today's pre-lecture content? Are other people in your zone also confused by this, or can someone explain it?
2. How is A0 going so far?

Tasks

- Pre-lecture homeworks: Due daily
 - (PL settings to be updated)
- Assignment 0: **Due today**
- Pre-course survey/textbook access–anyone not received?
- Signing up for Oral Exams?
Due yesterday–done?
Might be a room change–still basement
- Practice exam through the CBTF?
Required by Jul 2
(Schedule is down for today–ask around!)
- Project proposal **due Monday**

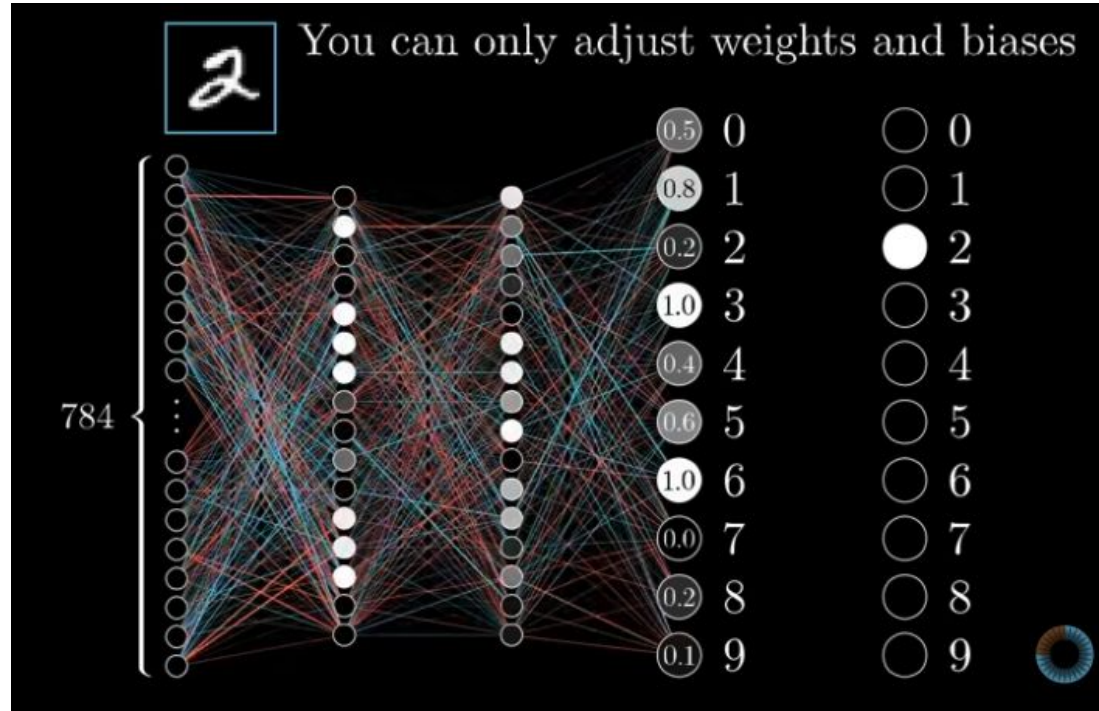
Upcoming

- **Assignment 1** will be posted later today
- **Project details** will be posted Thursday
 - Enough info to decide between options?
Kaggle: ARC dataset (multiple choice science questions)
 - Questions about the project?

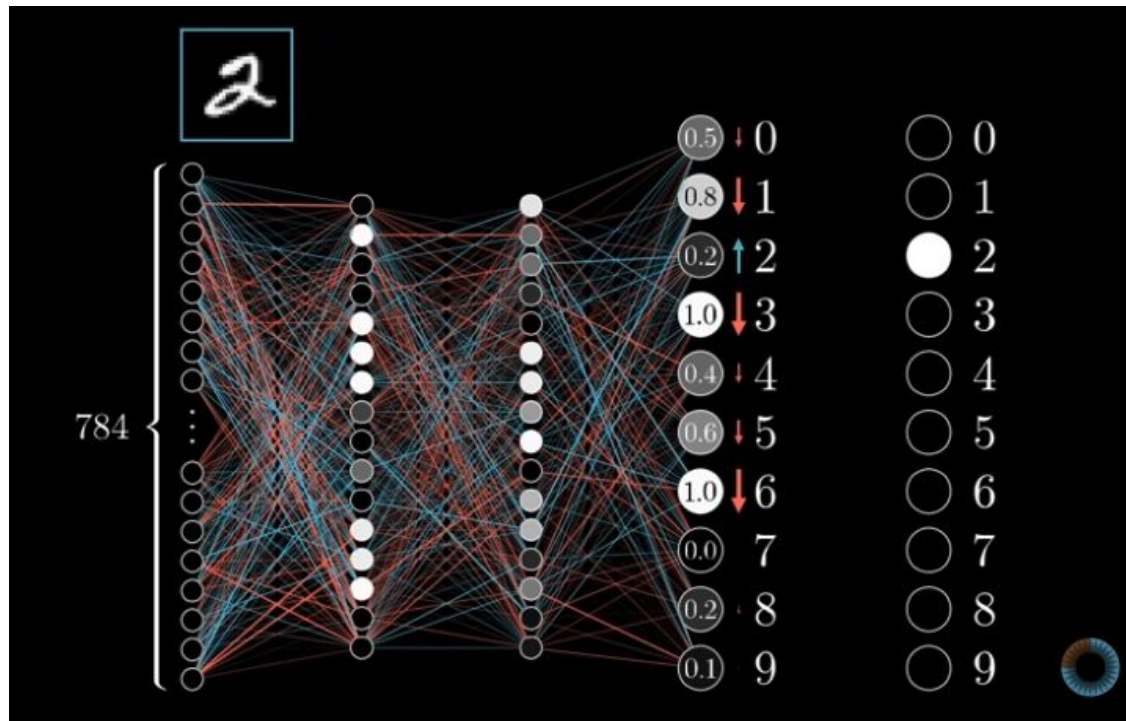
Where we are...

- We know why we want neural networks
 - They can make interesting, nonlinear, arbitrarily complicated decision boundaries
- We know about how they classify things once you give them an input (“forward pass”)
- But how do we determine the weights used for that forward pass?

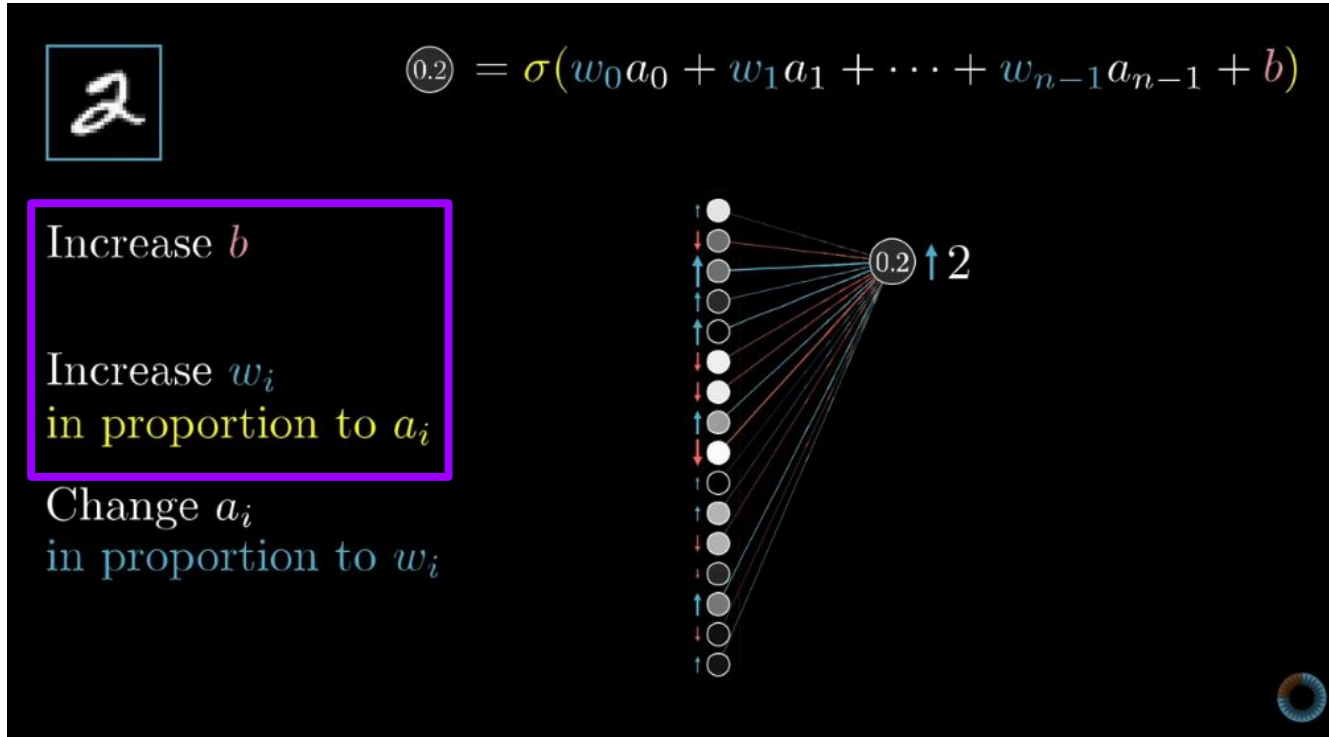
What is our goal? What are the rules?



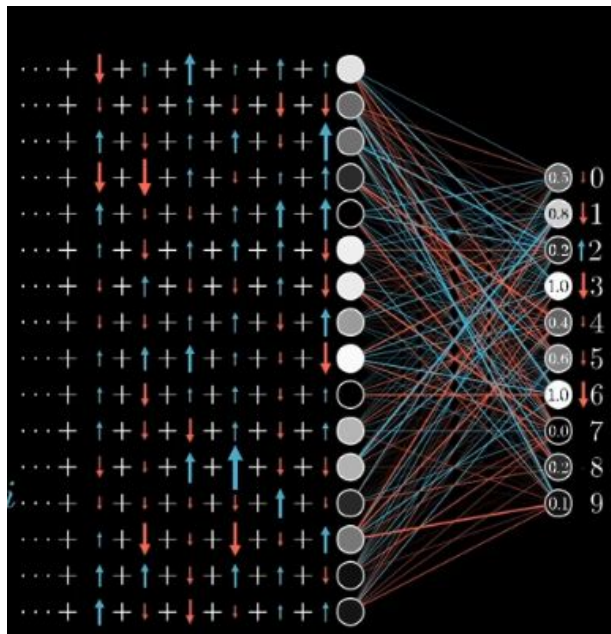
What do the arrows mean here? (Two attributes...)



The toggles we have for making those arrows happen:

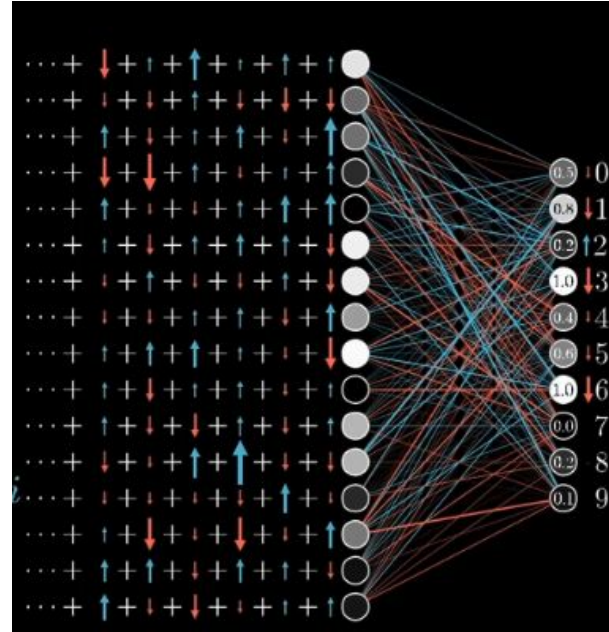


What's being summed here?



What's being summed here?

Each column is the “desires” of an output neuron



Let's zoom in on how we get those weight changes to happen.

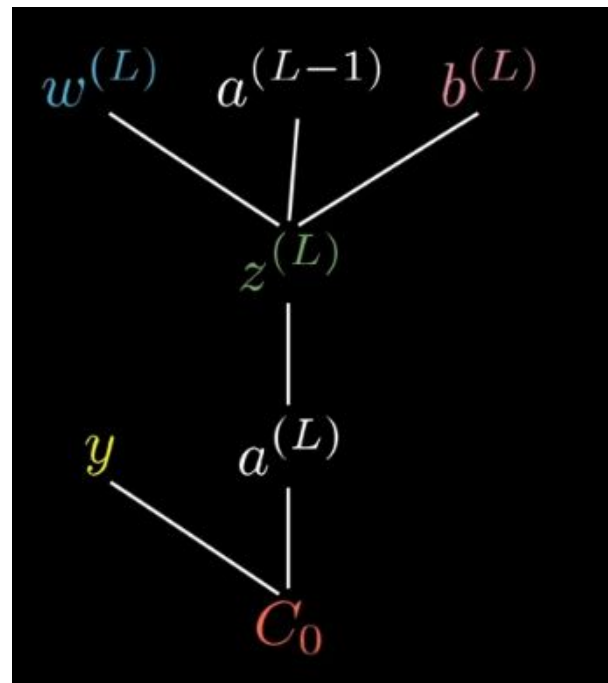
- Does this remind you of anything we learned about yesterday?
- Similarities/differences?

(fun fact)

Representational competence:

“A way of describing how a person uses a variety of perceptions of reality to make sense of and communicate understandings.”

(“Towards a Definition of Representational Competence”, 2018)



3b1b: Backpropagation, calculus

Let's zoom in on how we get those weight changes to happen.

- Suppose you have a loss function $L(f(x, y, z))$

$$f(x, y, z) = (x + y)z$$

where $x = -2$, $y = 5$, and $z = -4$.

(Right now this is an arbitrary function, but the same applies to a loss function, and x and y will be inputs)

- In training, we want to compute

$$\frac{\partial \mathcal{L}(f(x, y, z))}{\partial x}, \frac{\partial \mathcal{L}(f(x, y, z))}{\partial y}, \text{ and } \frac{\partial \mathcal{L}(f(x, y, z))}{\partial z}.$$

Let's zoom in on how we get those weight changes to happen.

Suppose we already know $\frac{\partial \mathcal{L}(f(x,y,z))}{\partial f} \equiv \frac{\partial \mathcal{L}(f)}{\partial f}$, e.g. say 1.0

Next we want to compute

$$\frac{\partial \mathcal{L}(f(x,y,z))}{\partial x} = \frac{\partial \mathcal{L}(f(x,y,z))}{\partial f} \frac{\partial f}{\partial x},$$

$$\frac{\partial \mathcal{L}(f(x,y,z))}{\partial y} = \frac{\partial \mathcal{L}(f(x,y,z))}{\partial f} \frac{\partial f}{\partial y},$$

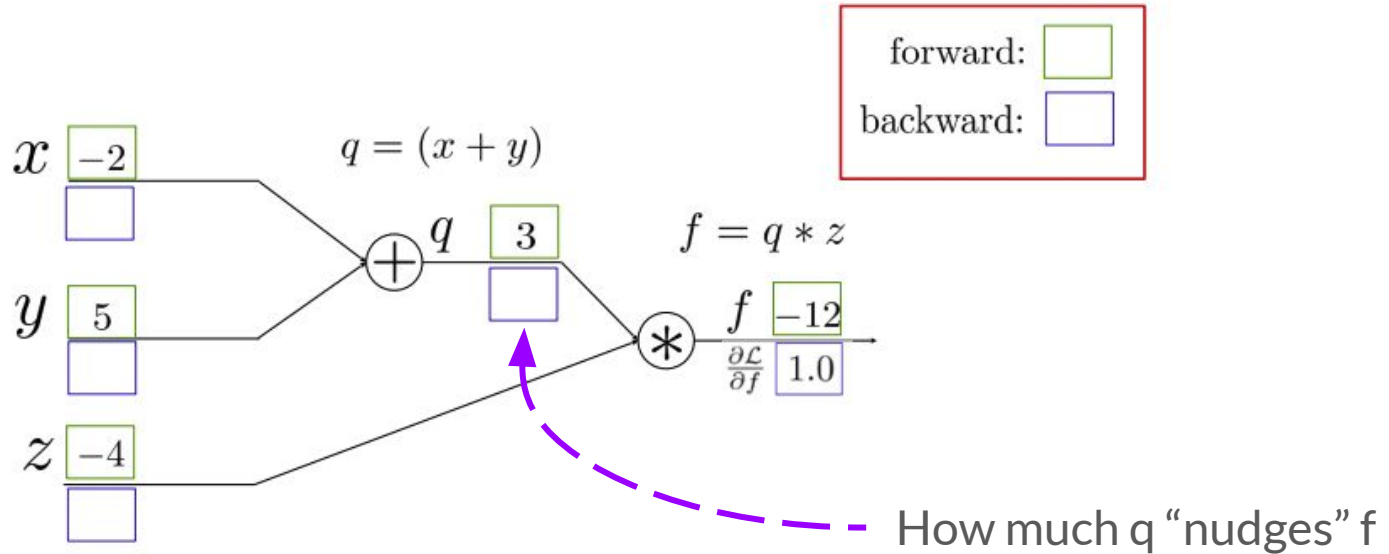
$$\frac{\partial \mathcal{L}(f(x,y,z))}{\partial z} = \frac{\partial \mathcal{L}(f(x,y,z))}{\partial f} \frac{\partial f}{\partial z},$$

~~(You can think of this as a base case...just taking the derivative with respect to itself)~~

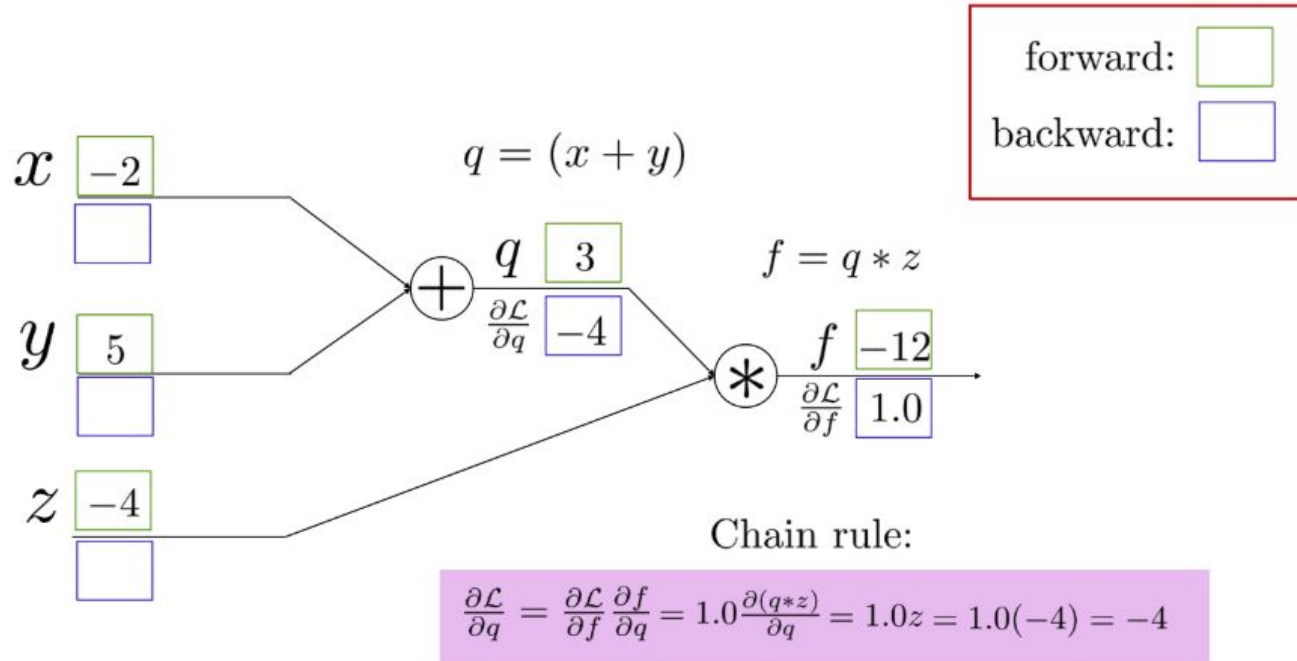
See slides after what we covered today for correction.

Yesterday: “Forward Pass”. Today: Backward Pass!

$$f(x, y, z) = (x + y)z \quad \text{where } x = -2, y = 5, \text{ and } z = -4.$$



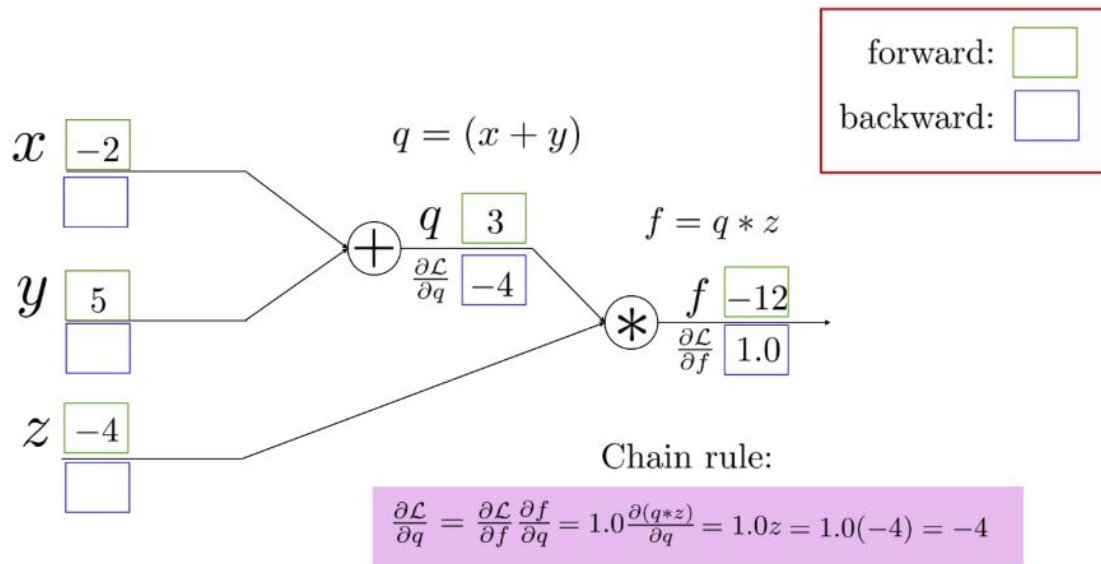
Backward Pass: $\frac{\partial \mathcal{L}}{\partial q} = \frac{\partial \mathcal{L}}{\partial f} \frac{\partial f}{\partial q}$



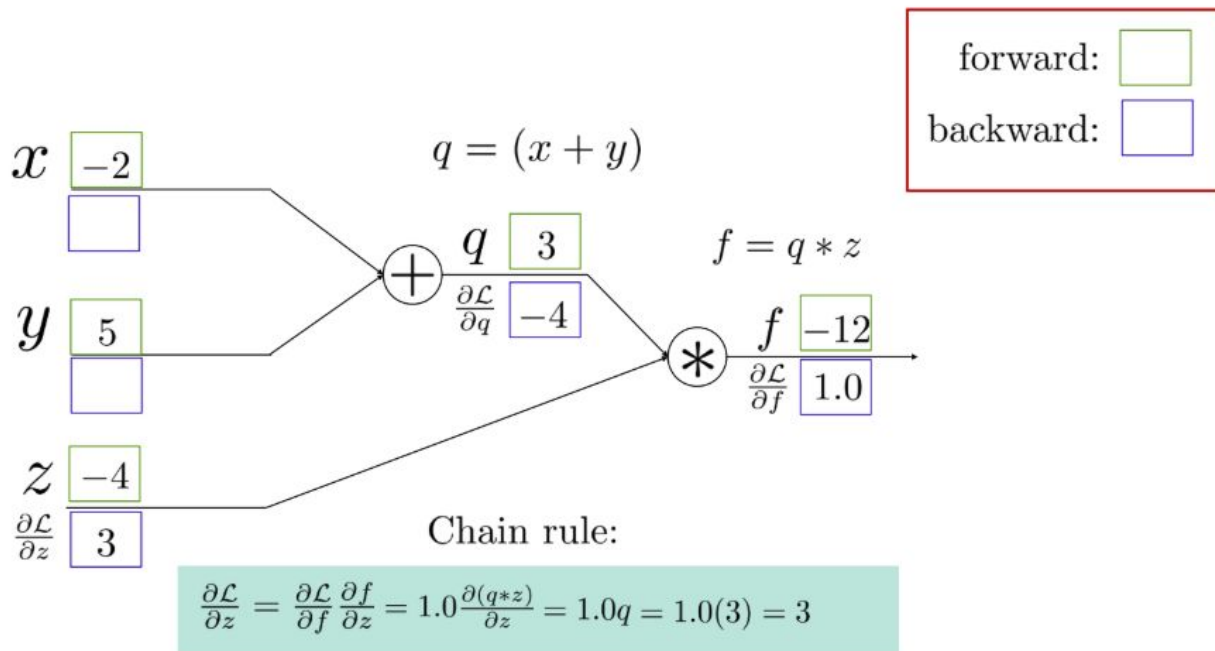
Your turn! What's next?

What we have so far:

- Fill **one** next step in under the edges from yesterday's graph
- When you've finished, compare with the person next to you



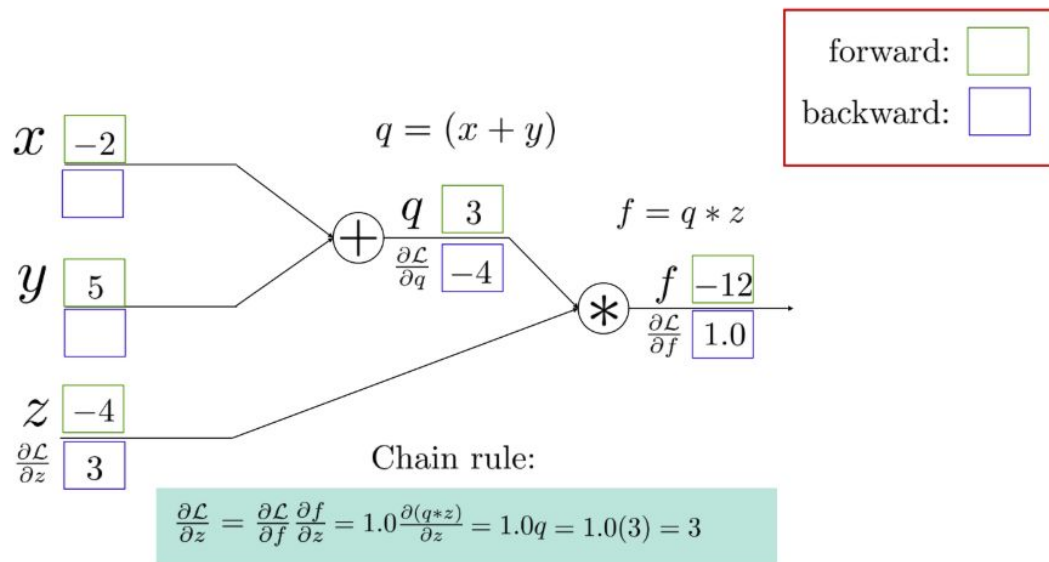
Backward Pass: $\frac{\partial \mathcal{L}}{\partial z} = \frac{\partial \mathcal{L}}{\partial f} \frac{\partial f}{\partial z}$



Keep going...

- Fill **one** next step in under the edges from yesterday's graph
- When you've finished, compare with the person next to you

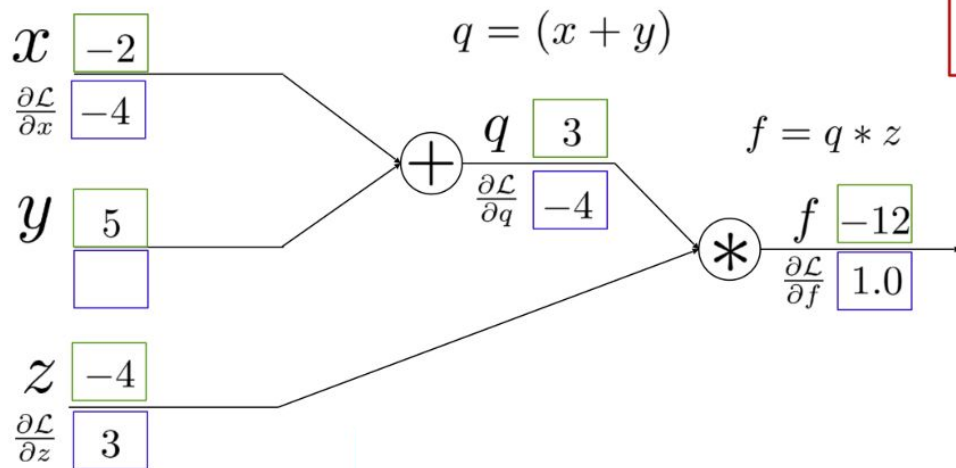
What we have so far:



Backward Pass: $\frac{\partial \mathcal{L}}{\partial x} = \frac{\partial \mathcal{L}}{\partial f} \frac{\partial f}{\partial q} \frac{\partial q}{\partial x} = \frac{\partial \mathcal{L}}{\partial q} \frac{\partial q}{\partial x}$

Chain rule:

$$\frac{\partial \mathcal{L}}{\partial x} = \frac{\partial \mathcal{L}}{\partial f} \frac{\partial f}{\partial q} \frac{\partial q}{\partial x} = \frac{\partial \mathcal{L}}{\partial q} \frac{\partial q}{\partial x} = (-4) \frac{\partial(x+y)}{\partial x} = (-4) * (1) = -4$$



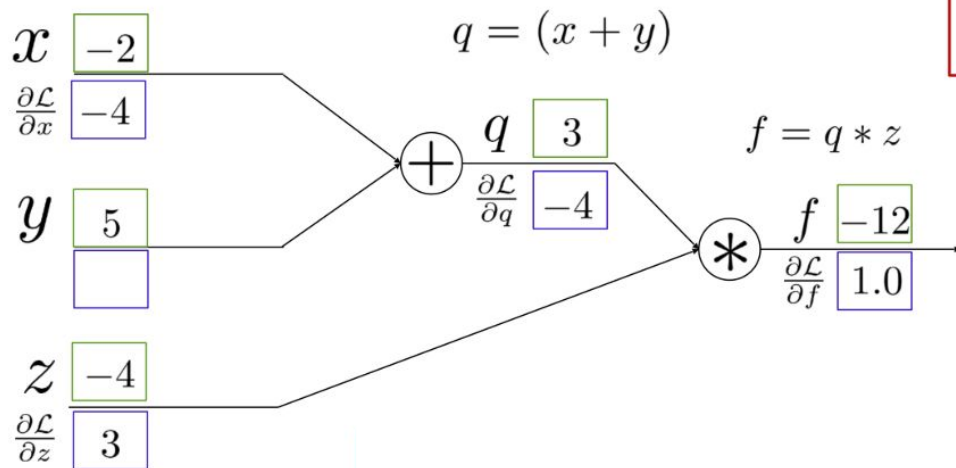
forward:

backward:

Backward Pass: $\frac{\partial \mathcal{L}}{\partial x} = \frac{\partial \mathcal{L}}{\partial f} \frac{\partial f}{\partial q} \frac{\partial q}{\partial x} = \frac{\partial \mathcal{L}}{\partial q} \frac{\partial q}{\partial x}$

Chain rule:

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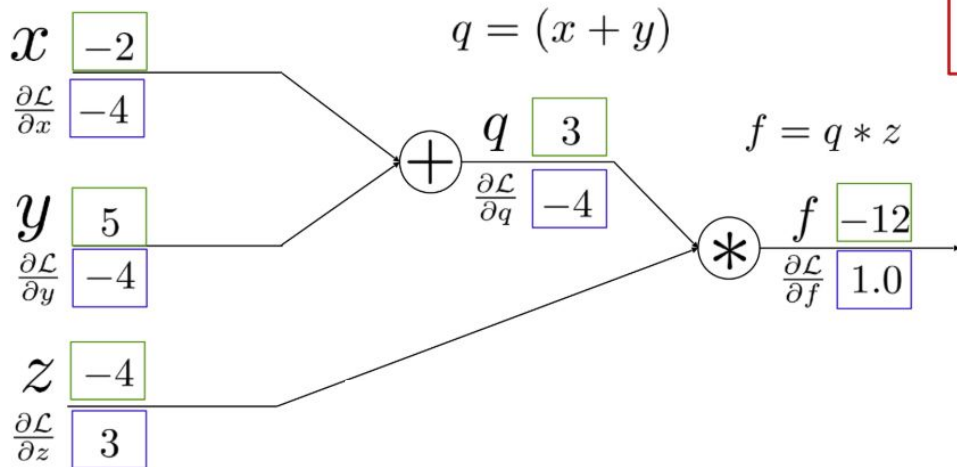
forward:

backward:

Backward Pass: $\frac{\partial \mathcal{L}}{\partial y} = \frac{\partial \mathcal{L}}{\partial f} \frac{\partial f}{\partial q} \frac{\partial q}{\partial y} = \frac{\partial \mathcal{L}}{\partial q} \frac{\partial q}{\partial y}$

Chain rule:

$$\frac{\partial \mathcal{L}}{\partial y} = \frac{\partial \mathcal{L}}{\partial f} \frac{\partial f}{\partial q} \frac{\partial q}{\partial y} = \frac{\partial \mathcal{L}}{\partial q} \frac{\partial q}{\partial y} = (-4) \frac{\partial (x+y)}{\partial y} = (-4) * (1) = -4$$



forward:

backward:

Congrats!

Break time. See you in 5 minutes!

Neural Networks & Human Cognition

Career [\[edit\]](#)

In 1986 McClelland published *Parallel Distributed Processing: Explorations in the Microstructure of Cognition*^[3] with [David Rumelhart](#), which some still regard as a bible for [cognitive scientists](#). [Geoffrey Hinton](#) was a member of the PDP group. McClelland's present work focuses on learning, memory processes, and [psycholinguistics](#), still within the framework of connectionist models. He is a former chair of the [Rumelhart Prize](#) committee, having collaborated with Rumelhart for many years, and himself received the award in 2010 at the Cognitive Science Society Annual Conference in Portland, Oregon.

- Our brains don't do backprop, but they do parallel distributed processing...
- Stay tuned! He'll be guest lecturing on Jul 20.

James Lloyd McClelland



Born	December 1, 1948 (age 77) Cambridge, Massachusetts
Alma mater	Columbia University University of Pennsylvania
Awards	Grawemeyer Award in Psychology (2002) Rumelhart Prize (2010)
Scientific career	
Fields	Psychology
Website	stanford.edu/~jlmcc/ ↗

Can you guess what problem we're doing next?

Suppose we have a loss function $L(f(w_0, x_0, w_1, x_1, w_2))$

$$f(w_0, x_0, w_1, x_1, w_2) = \frac{1}{1 + e^{-(w_0 x_0 + w_1 x_1 + w_2)}}$$

where $w_0 = 2$, $x_0 = -1$, $w_1 = -3$, $x_1 = -2$, and $w_2 = -3$.

In training, we want to compute the gradient

Which is...? Which terms do we care about? Why?

Can you guess what problem we're doing next?

Suppose we have a loss function $L(f(w_0, x_0, w_1, x_1, w_2))$

$$f(w_0, x_0, w_1, x_1, w_2) = \frac{1}{1 + e^{-(w_0 x_0 + w_1 x_1 + w_2)}}$$

where $w_0 = 2$, $x_0 = -1$, $w_1 = -3$, $x_1 = -2$, and $w_2 = -3$.

In training, we want to compute the gradient
and pick out the following terms:

$$\frac{\partial L}{\partial w_0} \quad \frac{\partial L}{\partial w_1} \quad \frac{\partial L}{\partial w_2}$$

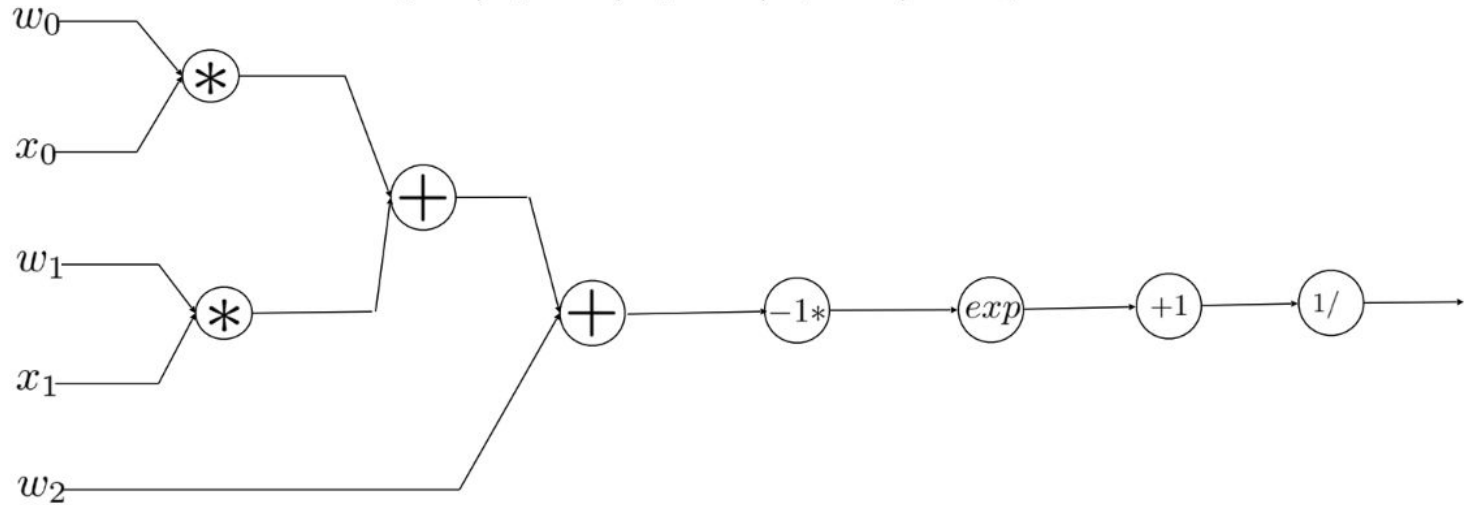
Only $\partial L / \partial w$ gets used for the weight update.

(Remember: only w is a toggle we have access to change...can't control our training data)

Yesterday: Forward Pass

$$f(w_0, x_0, w_1, x_1, w_2) = \frac{1}{1 + e^{-(w_0 x_0 + w_1 x_1 + w_2)}}$$

where $w_0 = 2$, $x_0 = -1$, $w_1 = -3$, $x_1 = -2$, and $w_2 = -3$.



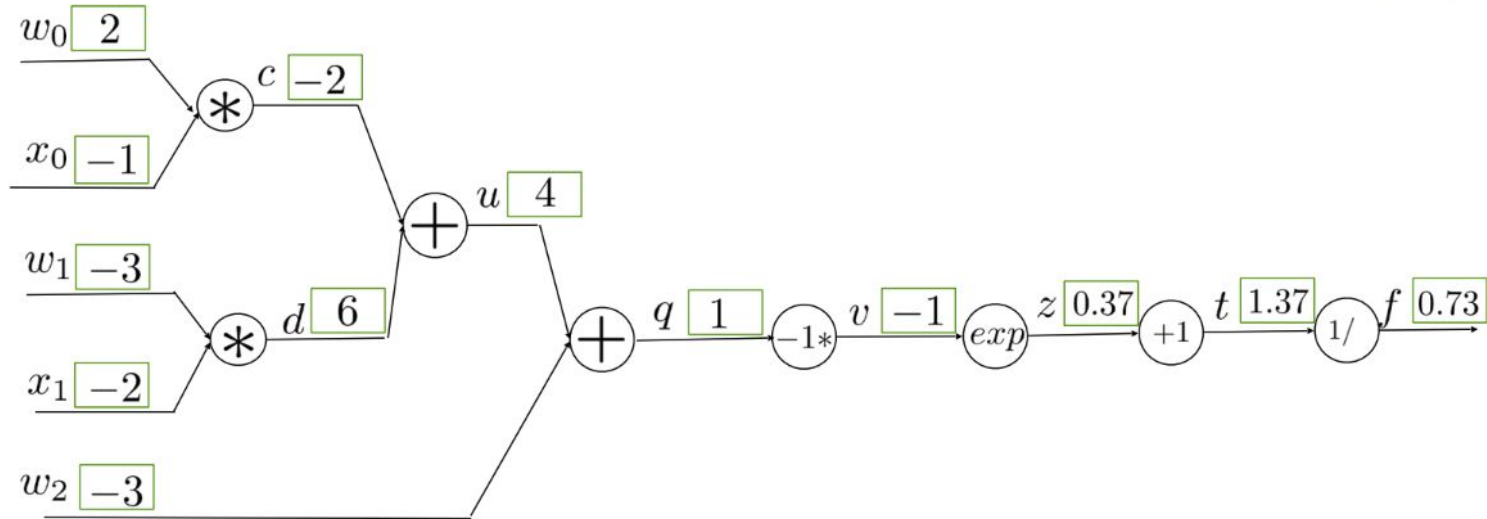
End of where we reached in class

Plugging in the inputs and starting weights

$$f(w_0, x_0, w_1, x_1, w_2) = \frac{1}{1 + e^{-(w_0 x_0 + w_1 x_1 + w_2)}}$$

where $w_0 = 2$, $x_0 = -1$, $w_1 = -3$, $x_1 = -2$, and $w_2 = -3$.

forward:

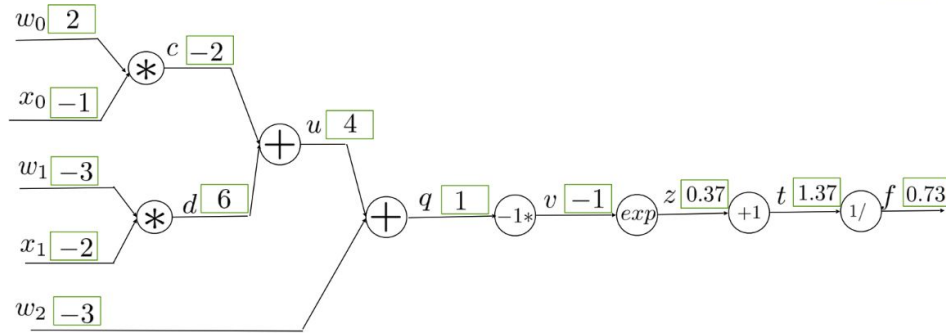


What are each of the functions in this graph?

$$f(w_0, x_0, w_1, x_1, w_2) = \frac{1}{1 + e^{-(w_0 x_0 + w_1 x_1 + w_2)}}$$

where $w_0 = 2$, $x_0 = -1$, $w_1 = -3$, $x_1 = -2$, and $w_2 = -3$.

forward:



Write what each function is for:

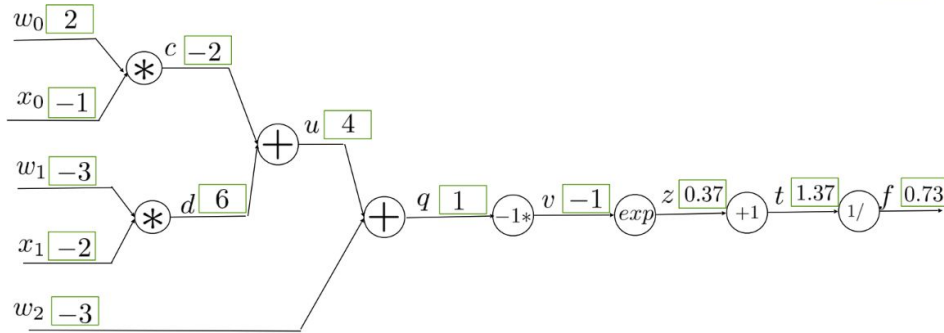
- f in terms of t
- t in terms of ...
- z ...
- v ...
- q ...
- u ...
- c ...
- d ...

Compare with the person next to you. Looking similar?

$$f(w_0, x_0, w_1, x_1, w_2) = \frac{1}{1 + e^{-(w_0 x_0 + w_1 x_1 + w_2)}}$$

where $w_0 = 2$, $x_0 = -1$, $w_1 = -3$, $x_1 = -2$, and $w_2 = -3$.

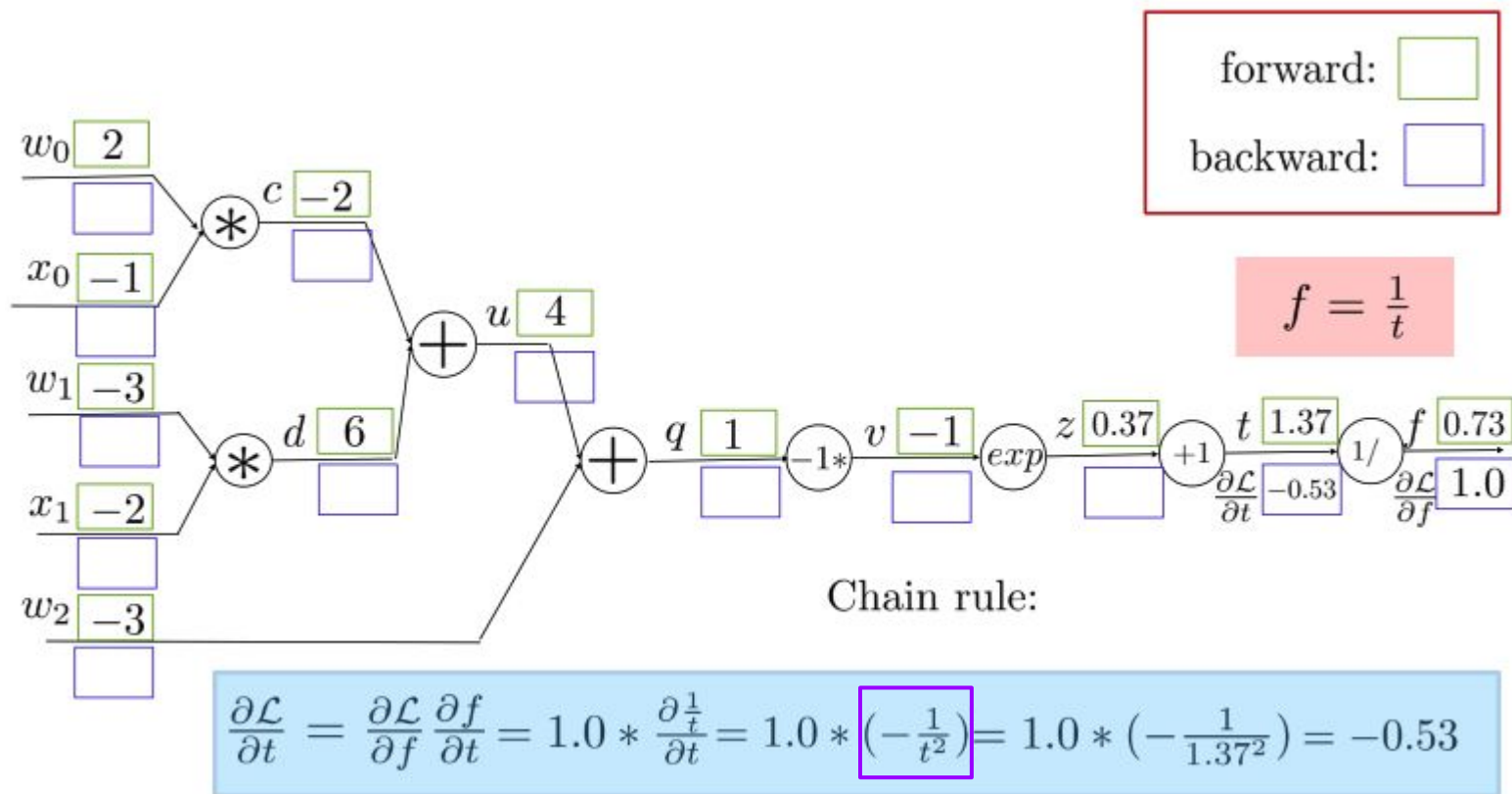
forward:



Write what each function is for:

- f in terms of t
- t in terms of ...
- z ...
- v ...
- q ...
- u ...
- c ...
- d ...

Backward pass

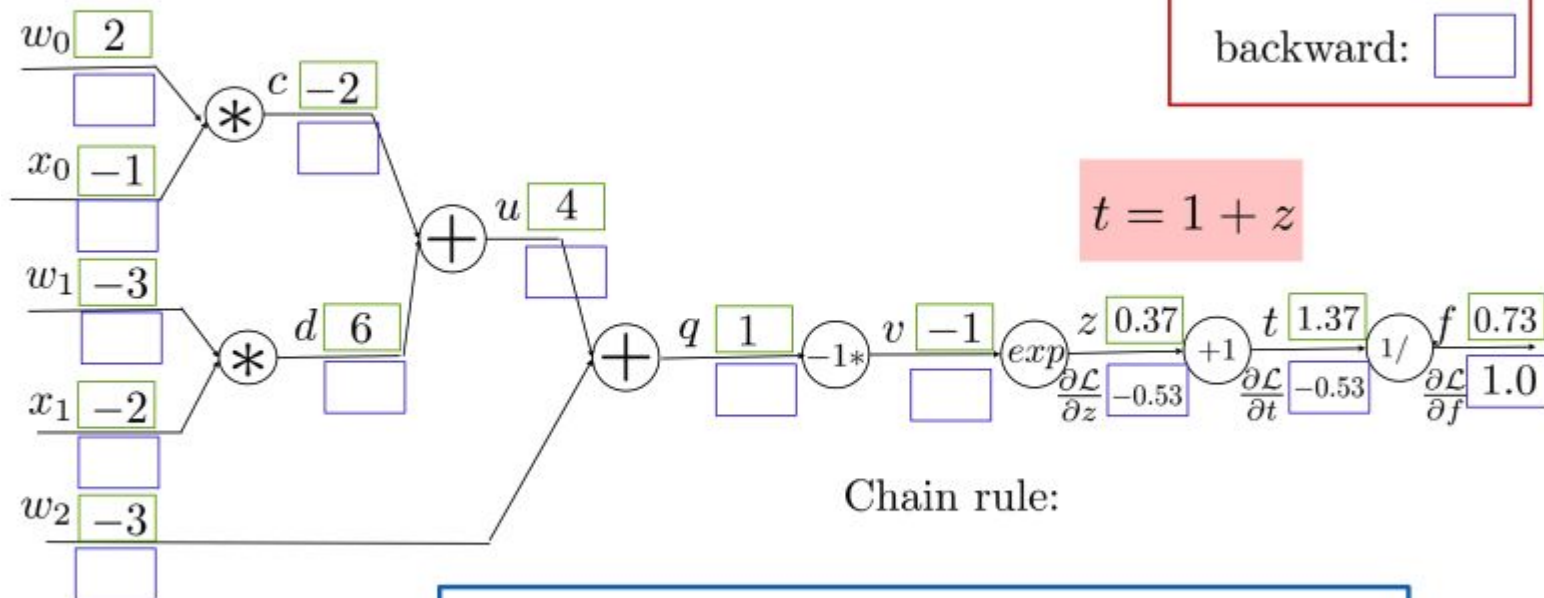


Backward pass

$$f(w_0, x_0, w_1, x_1, w_2) = \frac{1}{1 + e^{-(w_0 x_0 + w_1 x_1 + w_2)}}$$

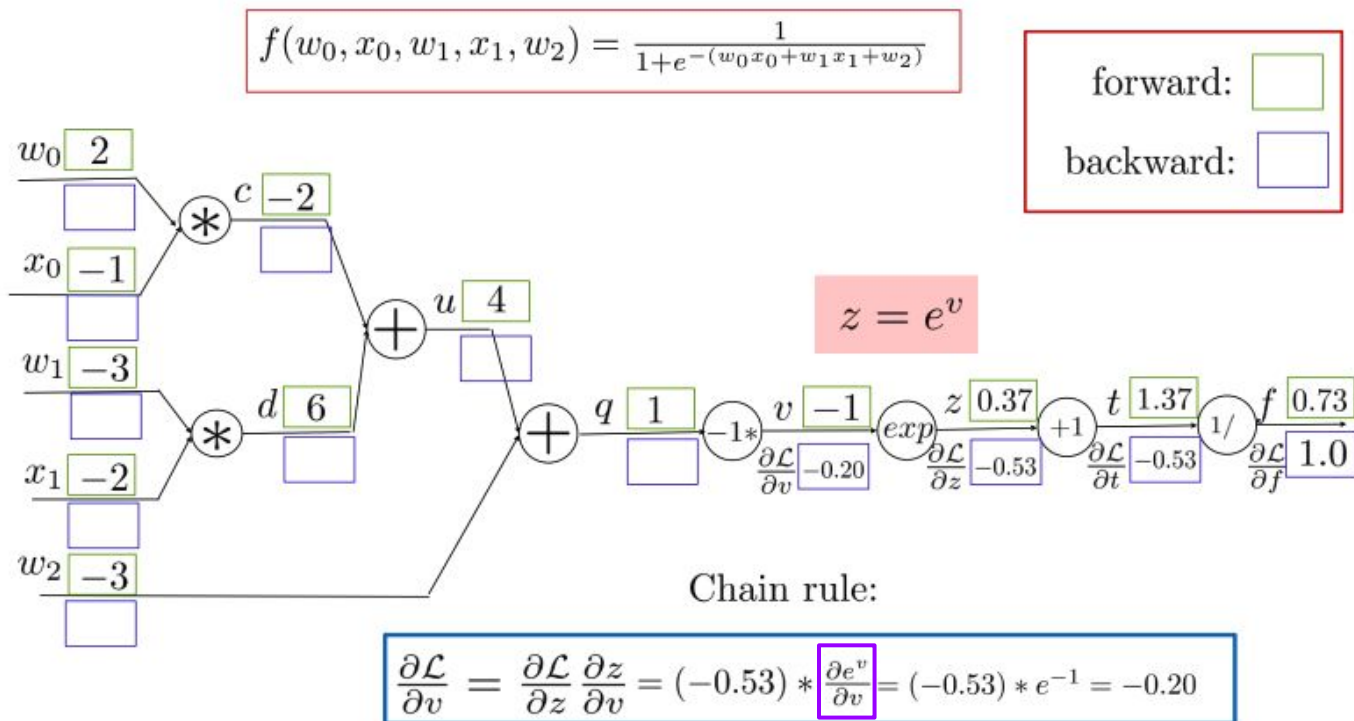
forward:

backward:

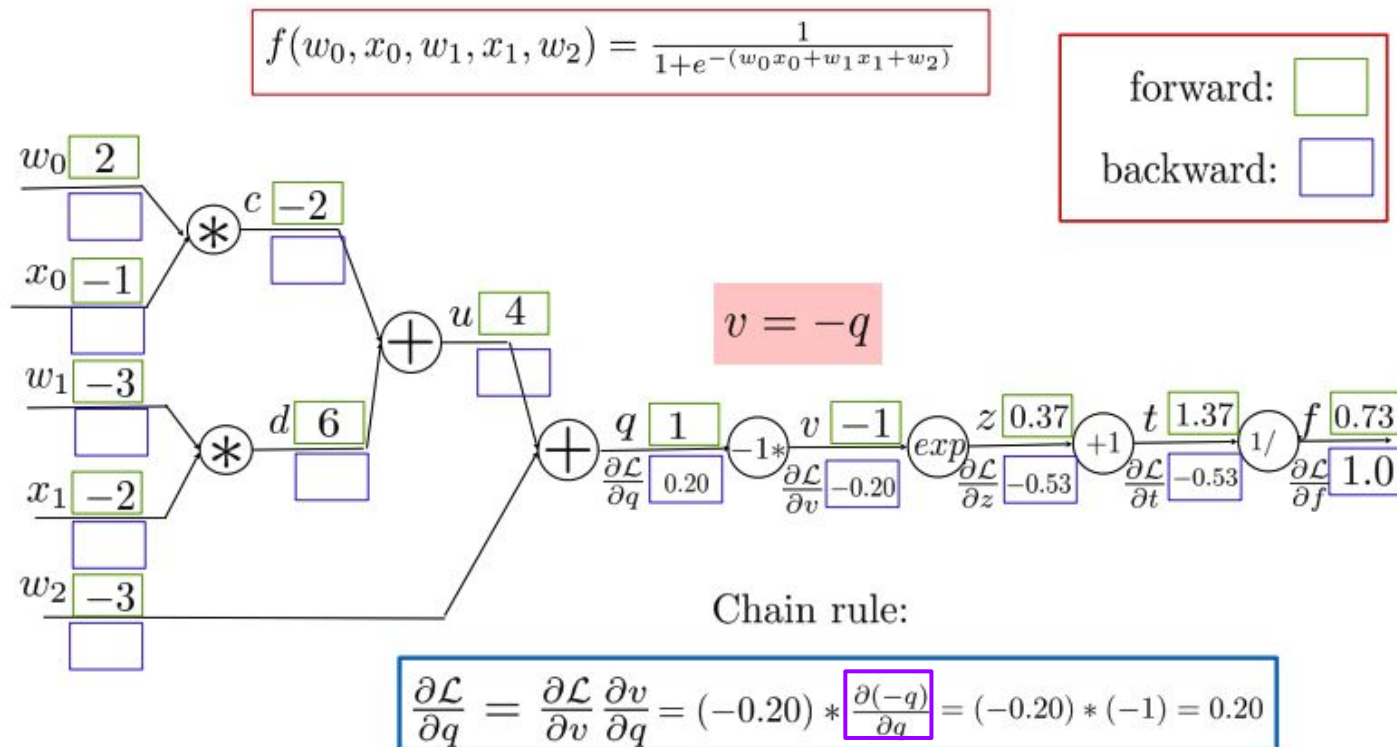


$$\frac{\partial \mathcal{L}}{\partial z} = \frac{\partial \mathcal{L}}{\partial t} \frac{\partial t}{\partial z} = (-0.53) * \frac{\partial(1+z)}{\partial z} = (-0.53) * 1 = -0.53$$

Backward pass



Backward pass

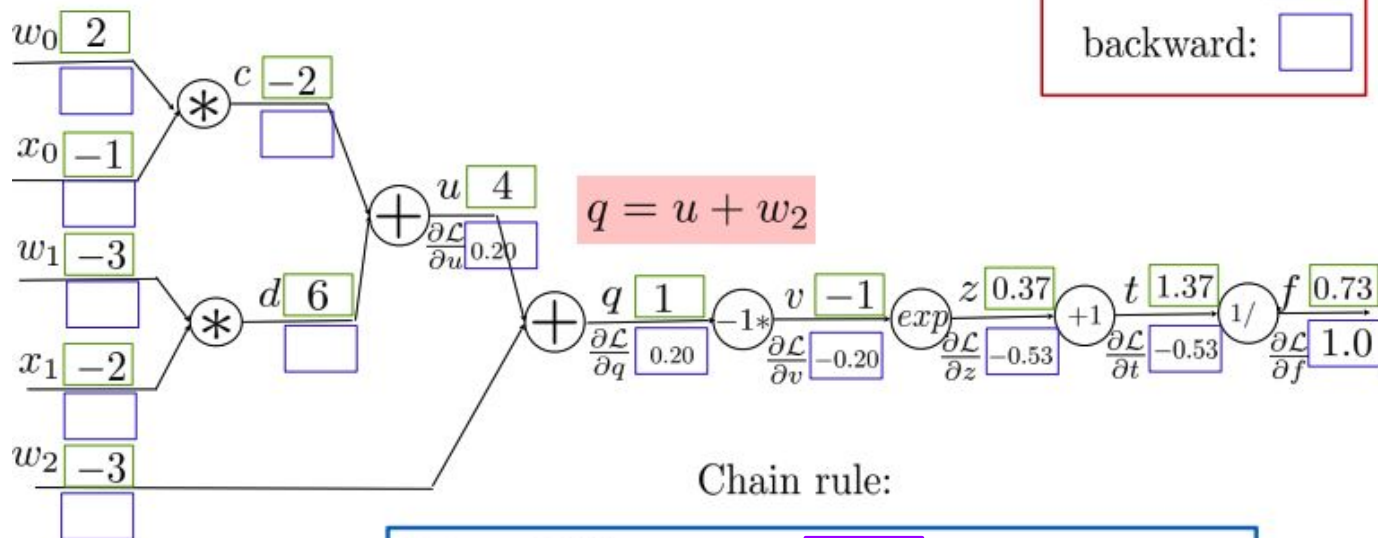


Backward pass

$$f(w_0, x_0, w_1, x_1, w_2) = \frac{1}{1 + e^{-(w_0 x_0 + w_1 x_1 + w_2)}}$$

forward:

backward:



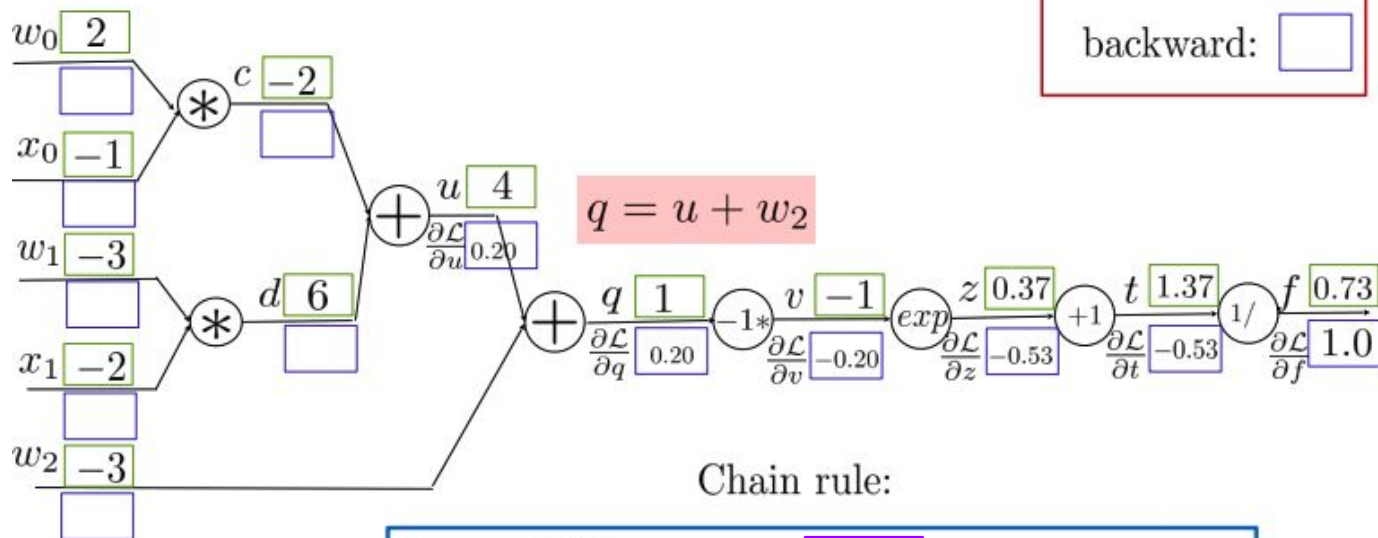
$$\frac{\partial \mathcal{L}}{\partial u} = \frac{\partial \mathcal{L}}{\partial q} \frac{\partial q}{\partial u} = (0.20) * \frac{\partial (u + w_2)}{\partial u} = (0.20) * (1) = 0.20$$

Backward pass

$$f(w_0, x_0, w_1, x_1, w_2) = \frac{1}{1 + e^{-(w_0 x_0 + w_1 x_1 + w_2)}}$$

forward:

backward:



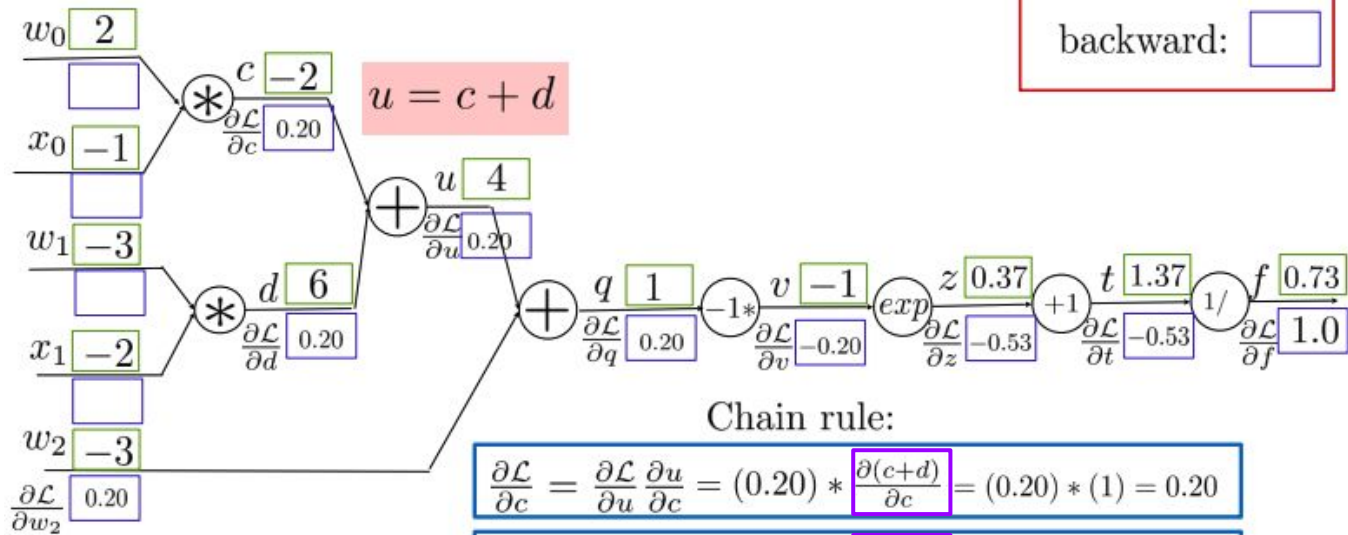
$$\frac{\partial \mathcal{L}}{\partial u} = \frac{\partial \mathcal{L}}{\partial q} \frac{\partial q}{\partial u} = (0.20) * \frac{\partial (u + w_2)}{\partial u} = (0.20) * (1) = 0.20$$

Backward pass

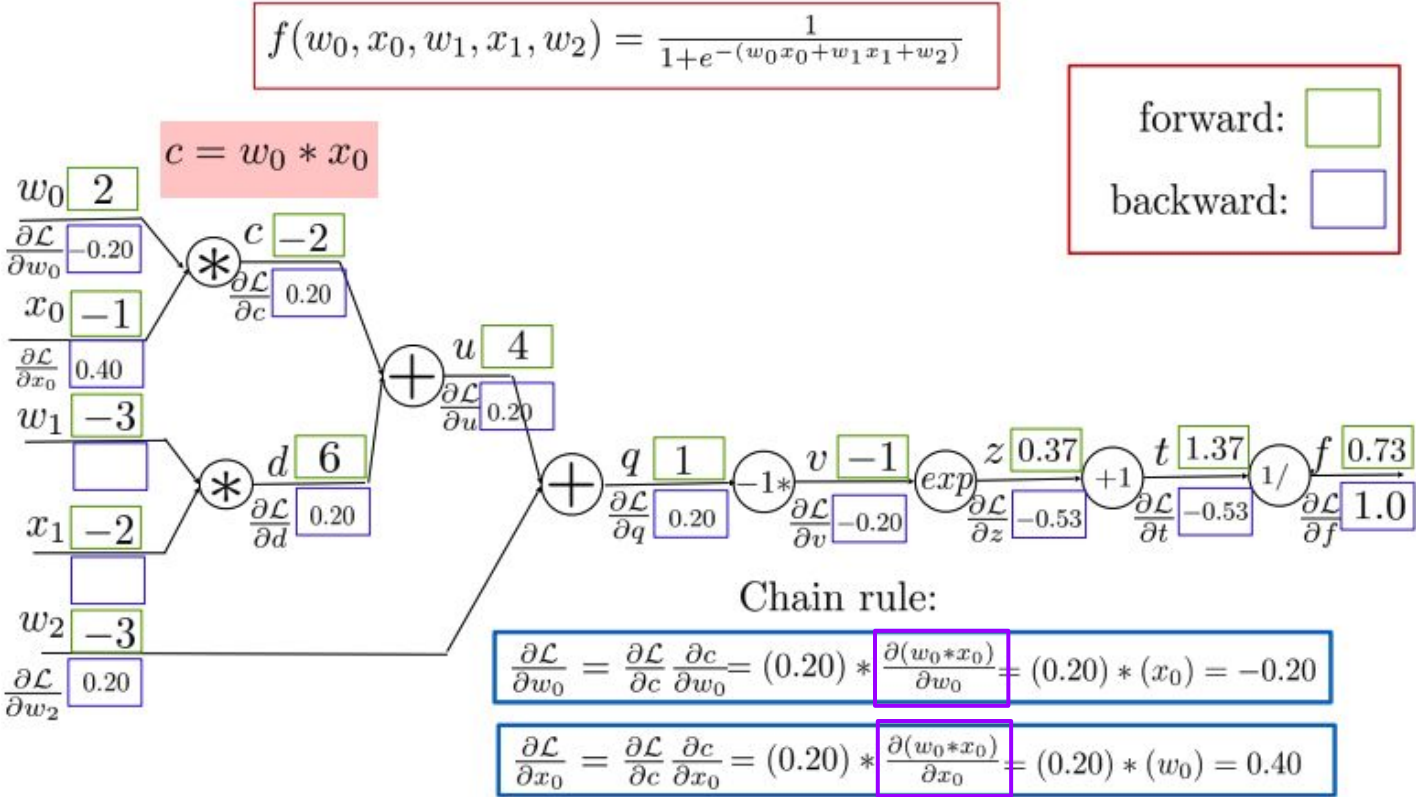
$$f(w_0, x_0, w_1, x_1, w_2) = \frac{1}{1 + e^{-(w_0 x_0 + w_1 x_1 + w_2)}}$$

forward:

backward:



Backward pass

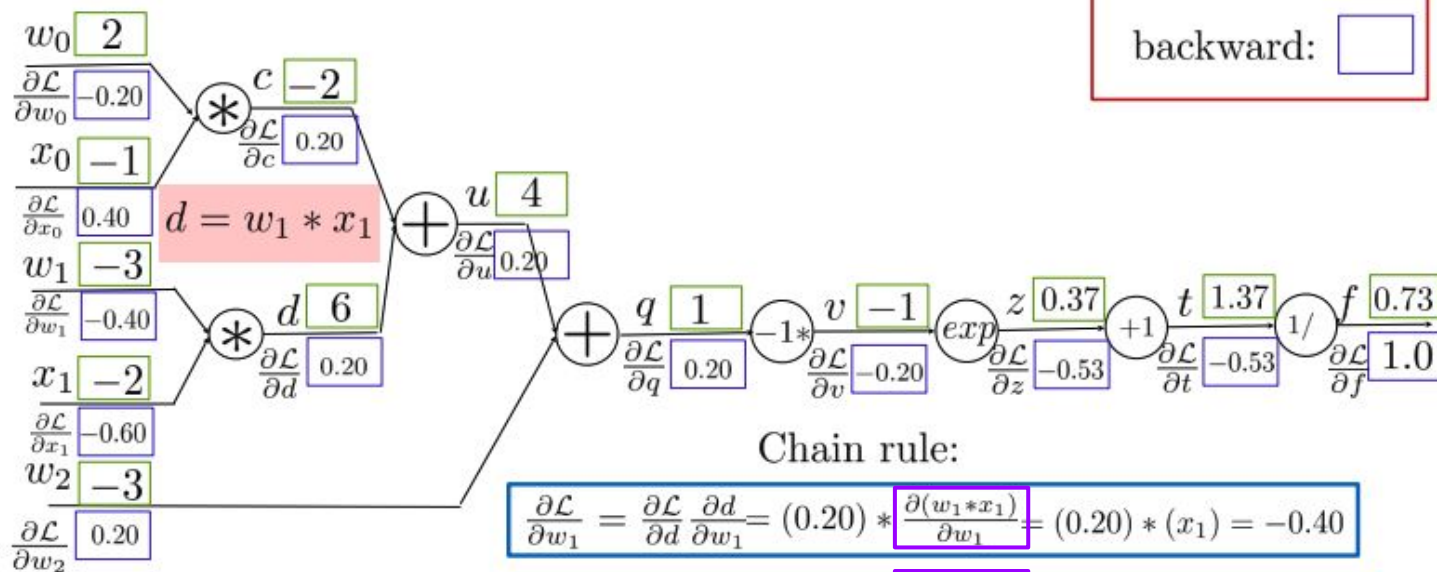


Backward pass

$$f(w_0, x_0, w_1, x_1, w_2) = \frac{1}{1 + e^{-(w_0 x_0 + w_1 x_1 + w_2)}}$$

forward:

backward:

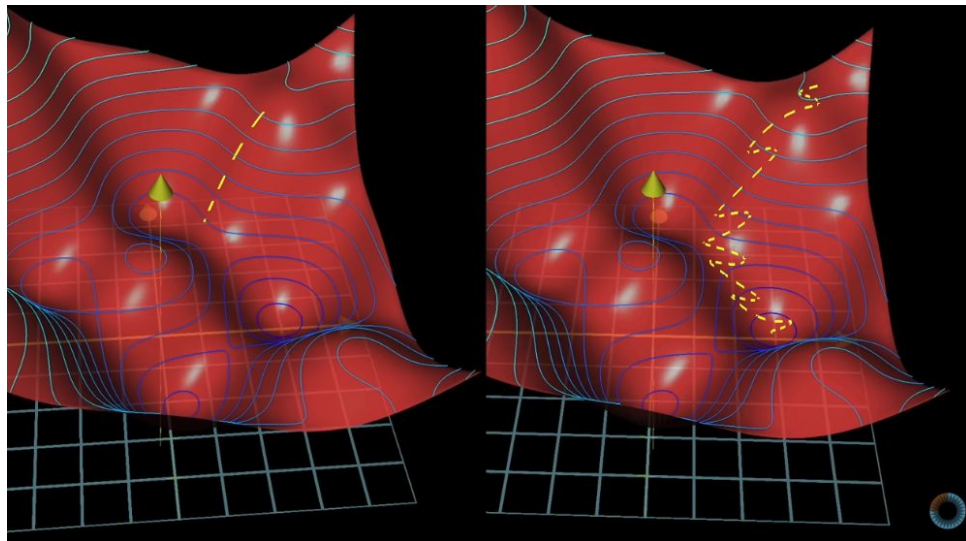


Congrats again!

- Questions?

Stochastic Gradient Descent

- Updating the network on every single training example is very computationally expensive
- Instead, SGD computes it on “minibatches” of the data



Jacobian? Gradient?

Both of these are about keeping track of derivatives in multiple dimensions.

- What's the difference between a gradient and a Jacobian? (Hint: shapes!)
- When do we use each of them in ML/DL?

Gradient

How did we communicate rate of change in 2D equations in single-variable calculus?

- A line $y=x^2$ has a derivative of $2x$. We just have the one expression, $2x$.

What do we do when we have two dimensions to account for?

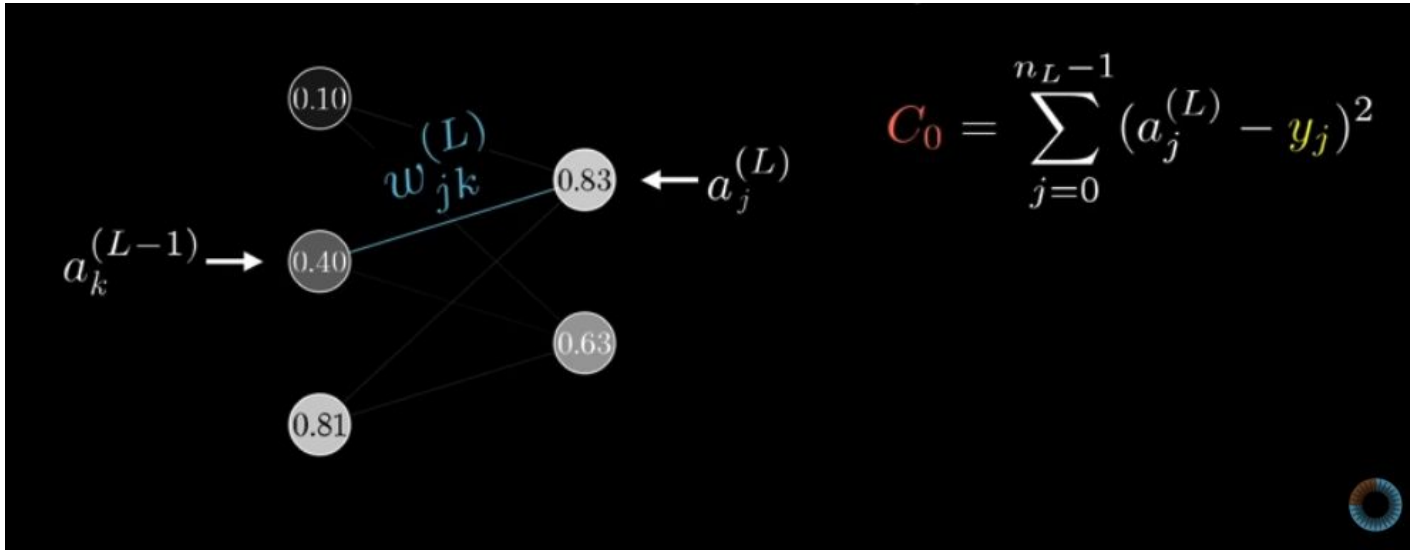
- Gradient! Two expressions we store in a vector: Derivative w.r.t. x is $3x$, w.r.t y is y . This communicates what we wanted: our vector below tells us exactly how much the gradient changes with each a given change in x and y .

$$\nabla f(x, y) = (3x, y)$$

- A vector is just a way of storing values in multiple dimensions. All the math is the same as what you first learned in calculus; vectors are a helpful way to keep track.

Jacobian

- A list of all the gradients you're computing in backprop.



Updates/corrections to points from class

- In the last step of the computation graph when we talked about dL/df , I said that it is always 1 because they are the same. This is incorrect.
 - The reason that dL/df is often set to 1 in computational graph examples is simply so that we don't need to decide on or get into the details of any particular loss function (L) in order to proceed with the backprop computation.
- I said that you don't need the model's output in order to backprop through the computation graph. This is incorrect—and quite important!
 - dL/df is in fact the derivative of the loss with respect to the output.